

The Thue-Morse Sequence

Fall 2025 ARML Power Contest

The Solutions

1. The perfect powers less than 100 are $\boxed{1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, \text{ and } 81}$.
2. Here are a few such combinations:

- $1 + 16 + 64 - 81$
- $1 - 16 - 49 + 64$
- $1 + 27 + 36 - 64$
- $8 + 16 + 25 - 49$
- $32 + 49 - 81$

This list is not intended to be exhaustive.

3. The squares could all be multiplied out and like terms collected to obtain the result (that is, “bash it”). A more general approach is to do it symbolically. Let σ_k be the sign (that is, plus or minus) of k^2 in the original expression. Then this question is asking about the sum

$$\begin{aligned} \sum_{k=1}^8 \sigma_k (a-1+k)^2 &= \sum_{k=1}^8 \sigma_k ((a-1)^2 + 2k(a-1) + k^2) \\ &= \sum_{k=1}^8 \sigma_k (a-1)^2 + \sum_{k=1}^8 2\sigma_k k(a-1) + \sum_{k=1}^8 \sigma_k k^2. \end{aligned}$$

In the first sum, $(a-1)^2$ may be factored out of the sum, leaving the sum of σ_k of which four are positive and four are negative, so they all cancel each other. In the second sum, $2(a-1)$ may be factored out, leaving $1 - 2 - 3 + 4 - 5 + 6 + 7 - 8$, which clearly adds to zero. The third sum is simply the original sum given in the problem which is already known to add to zero.

4. $W_4 = 01101001\overline{01101001} = \boxed{0110100110010110}$.
5. This is a straightforward induction. $W_0 = 0$ has length $2^0 = 1$ by inspection. Then given that W_n has length 2^n , W_{n+1} is constructed by appending to W_n its complement, which has the same length, so the length of W_{n+1} is twice that of W_n , and $2 \cdot 2^n = 2^{n+1}$.
6. W_4 is given above and has length 16. Its last term is $s_{15} = 0$. Another 8 terms past this is the opposite of the eighth entry (careful—it’s labeled s_7). The opposite of 1 is $s_{23} = \boxed{0}$.
7. First, note that if a word W is a palindrome, so is $\overline{W}$. Second, note that $W_0 = 0$ is a palindrome. Now assume that W_{2k} is a palindrome for some k . Then

- $W_{2k+1} = W_{2k} \overline{W_{2k}}$. The final symbol of W_{2k+1} is the complement of the final symbol of W_{2k} . Since W_{2k} is a palindrome, its first and last symbols are the same. Thus the final symbol of W_{2k+1} is the complement of that word's first symbol, and thus W_{2k+1} cannot be a palindrome.
- $W_{2(k+1)} = W_{2k+2} = W_{2k+1} \overline{W_{2k+1}} = W_{2k} \overline{W_{2k}} \overline{W_{2k} \overline{W_{2k}}} = W_{2k} \overline{W_{2k}} \overline{W_{2k}} \overline{\overline{W_{2k}}}$. For any word W , $\overline{\overline{W}} = W$. Thus $W_{2k+2} = W_{2k} \overline{W_{2k}} \overline{W_{2k}} W_{2k}$ and since W_{2k} is a palindrome, so is this combination.

Taken together, these computations inductively show that $n = 2k$ is even, then W_n is a palindrome while if $n = 2k + 1$ is odd, it cannot be a palindrome.

8. There are a number of proofs that can work; the one shown here is a clever ploy that has very useful applications such as a nice proof of the general AM-GM inequality. The proof is by induction, but the induction variable will be in an exponent!

For a base case, 0 has no ones in its binary expansion, and $s_0 = 0$. Thus, the statement is true for all numbers less than 2^n when $n = 0$.

Now assume that for some n , the statement is true for all numbers less than 2^n . For any k with $2^n \leq k < 2^{n+1}$ write $k = 2^n + b$ where $b = k - 2^n$ is a nonnegative integer less than 2^n . The binary expansion for k is thus 1 followed by the binary expansion for b . The number of ones in the binary expansion for k is one greater than the number of ones in the expansion of b . The number of ones in the expansion for k is therefore odd if that of b is even and even if the expansion of b has an odd number of ones in it. Since $W_{n+1} = W_n \overline{W_n}$ the term $s_k = s_{2^n+b} = \overline{s_b}$ the induction is now complete: given that the statement is true for all numbers less than 2^n it will be true for all numbers less than 2^{n+1} . Since every nonnegative integer is less than some power of two, the statement is true for all nonnegative integers.

9. Another clever variation on induction is to prove two things simultaneously, so that each of the two can assist in proving the other. For this question, the base cases are $\mu(W_0) = \mu(0) = 01 = W_1$ and $\mu(\overline{W_0}) = \mu(1) = 10 = \overline{W_1}$.

Now assume that for some $k \geq 0$ both $\mu(W_k) = W_{k+1}$ and $\mu(\overline{W_k}) = \overline{W_{k+1}}$. Then, by definition, $\mu(W_{k+1}) = \mu(W_k \overline{W_k})$. By the inductive assumption, this becomes $W_{k+1} \overline{W_{k+1}}$ which, again by definition, equals $\overline{W_{k+2}}$. A similar computation swapping every word with its complement shows that $\mu(\overline{W_{k+1}}) = \overline{W_{k+2}}$. Since truth of the statement for k (for both W_k and $\overline{W_k}$) implies that for $k + 1$, the induction is complete.

10. This can be most easily shown as a consequence of the result proven in problem 8. Since s_k is the number of ones in the binary expansion of k , and the binary expansion of $2k$ is obtained from that of k by appending a zero to the right end of the digits, the number of ones in both cases is the same. That is, $s_{2k} = s_k$. On the other hand, the binary expansion of $2k + 1$ is obtained by appending a one to the end, so the number of ones changes from even to odd or vice versa. Thus, $s_{2k+1} = \overline{s_k} = 1 - s_k$.

11. (a) $\Delta_a p(x) = p(x) - p(x+a) = c_0 - c_0 = 0$. Note that this is also true when $c_0 = 0$, meaning the polynomial is the zero polynomial which technically is not degree zero.
- (b) Work by induction. For $d = 1$, the polynomial $p(x)$ will be of the form $p(x) = c_1x + c_0$ where $c_1 \neq 0$. Now

$$\Delta_a p(x) = p(x) - p(x+a) = (c_1x + c_0) - (c_1(x+a) + c_0) = -c_1a.$$

Note that neither c_1 nor a can be zero by the conditions of the problem. So $\Delta_a p(x)$ is a nonzero constant, which is the definition of a polynomial of degree 0. Now assume the statement is true for any polynomial of degree less than d . Rewrite $p(x) = c_dx^d + p^*(x)$ where $p^*(x)$ consists of all the terms of $p(x)$ of degree less than d , if any. Now notice that $\Delta_a p(x) = \Delta_a c_dx^d + \Delta_a p^*(x)$. Since $p^*(x)$ has degree less than d (or is the zero polynomial), $\Delta_a p^*(x)$ is a polynomial of degree at most $d-2$. On the other hand, $\Delta_a c_dx^d = c_d(x^d - (x+a)^d)$. Using the binomial theorem, this can be expanded as $c_d(x^d - x^d - dax^{d-1} - (\text{terms of smaller degree}))$. Since, by hypothesis, none of c_d , d , nor a are zero, the coefficient of x^{d-1} in $\Delta_a p(x)$ is nonzero, so the polynomial has degree exactly $d-1$.

- (c) Note that the case where some $a_i = 0$ must be considered, though it is very simple: $\Delta_0 p(x) = p(x) - p(x+0) = 0$. Further applications of Δ_a for any a —zero or nonzero—continue to produce the zero polynomial from this point onward.

On the other hand, if none of the a_i are zero, then by the previous part, each of the first d applications of Δ each reduce the degree by exactly one, so the result will be a polynomial of degree 0—a constant. Then the first part of this problem shows that Δ_{a_1} applied to this constant will yield zero.

12. Another induction argument. The base case was shown in the text preceding the problem. So proceed with the inductive step. Assume the statement is true for some k . Then

$$\begin{aligned} p_{k+1}(x) &= \Delta_{2^k} p_k(x) = p_k(x) - p_k(x + 2^k) \\ &= \sum_{j=0}^{2^k-1} (-1)^{s_j} p(x+j) - \sum_{j=0}^{2^k-1} (-1)^{s_j} p(x+j+2^k) \end{aligned}$$

Now note that when $0 \leq j \leq 2^k - 1$ then by definition s_j is the $j+1$ entry of W_k . Also note that s_{j+2^k} will be the complement of s_j because $j+2^k < 2^{k+1}$ so this s is the $j+2^k+1$ entry of $W_{k+1} = W_k \overline{W_k}$. Finally, note that raising -1 to the power of the complement of s is simply swapping $(-1)^0 = 1$ for $(-1)^1 = -1$ or vice versa. Putting

these together shows $-(-1)^{s_j} = (-1)^{\overline{s_j}} = (-1)^{s_{j+2^k}}$. Inserting this into the sum above,

$$\begin{aligned}
p_{k+1}(x) &= \sum_{j=0}^{2^k-1} (-1)^{s_j} p(x+j) + \sum_{j=0}^{2^k-1} (-1)^{s_{j+2^k}} p(x+j+2^k) \\
&= \sum_{j=0}^{2^k-1} (-1)^{s_j} p(x+j) + \sum_{j=2^k}^{2^{k+1}-1} (-1)^{s_j} p(x+j) \\
&= \sum_{j=0}^{2^{k+1}-1} (-1)^{s_j} p(x+j)
\end{aligned}$$

and the induction is complete.

13. The result follows from the previous problems applied to the polynomial $p(x) = x^d$ for each $d \leq n$. Thus, if $p(x) = x^d$, then by the result of the previous problem, $p_k(x) = \sum_{j=0}^{2^k-1} (-1)^{s_j} (x+j)^d$. This is the sum of 2^k consecutive d -th powers, with signs determined by the Thue-Morse sequence. By definition, $p_k(x)$ is the result of applying k Δ operations to $p(x)$, so by the result of 11(c) as long as $k \geq d+1$ the result will be zero. That is, 2^{n+1} consecutive d -th powers, combined with signs determined by the Thue-Morse sequence, yields zero as long as $n \geq d$, which is the desired result.
14. The number of cubes in a set is equal to the sum of the zeroth powers of the lengths of their edges. The total length of all edges of the cubes in a set is 12 times the sum of the lengths of the cubes' edges. The total surface area is 6 times the sum of the squares of the lengths of the edges, and the total volume is the sum of the cubes of the lengths of the edges. Since $16 = 2^4$, the previous result gives

$$\begin{aligned}
1^0 - 2^0 - 3^0 + 4^0 - 5^0 + 6^0 + 7^0 - 8^0 - 9^0 + 10^0 + 11^0 - 12^0 + 13^0 - 14^0 - 15^0 + 16^0 &= 0 \\
1^1 - 2^1 - 3^1 + 4^1 - 5^1 + 6^1 + 7^1 - 8^1 - 9^1 + 10^1 + 11^1 - 12^1 + 13^1 - 14^1 - 15^1 + 16^1 &= 0 \\
1^2 - 2^2 - 3^2 + 4^2 - 5^2 + 6^2 + 7^2 - 8^2 - 9^2 + 10^2 + 11^2 - 12^2 + 13^2 - 14^2 - 15^2 + 16^2 &= 0 \\
1^3 - 2^3 - 3^3 + 4^3 - 5^3 + 6^3 + 7^3 - 8^3 - 9^3 + 10^3 + 11^3 - 12^3 + 13^3 - 14^3 - 15^3 + 16^3 &= 0
\end{aligned}$$

So put the cubes of edge lengths 1, 4, 6, 7, 10, 11, 13, and 16 in one subset, and the rest in the other subset.

15. (a) Note that $W_0 = 0$, $W_1 = 01$, and $W_2 = 0110$, none of which contain either 000 or 111. Neither do their complements $\overline{W_0}$, $\overline{W_1}$, and $\overline{W_2}$. Note that if W_k does not contain 000 as a subword, then $\overline{W_k}$ cannot contain 111, and vice versa. Next, note that W_n for $n \geq 1$ starts with 01, and $\overline{W_n}$ with 10. Because W_{2k} is a palindrome, it ends with 10, while W_{2k+1} ends with the final two terms of $\overline{W_{2k}}$ which are 01. Thus W_n and $\overline{W_n}$ end in either 01 or 10 for any n . Now proceed by induction. The base cases are given above. Assume neither W_n nor $\overline{W_n}$ contain 000 or 111. Then $W_{n+1} = W_n \overline{W_n}$. This cannot contain 000 or 111 because either the subword would have to be totally contained in the beginning

W_n part or the ending $\overline{W_n}$ part—which is ruled out by the inductive assumption—or it would have to be “in the middle” where W_n and $\overline{W_n}$ meet—which is ruled out because the previous paragraph showed the neither the first two or last two terms of W_n or $\overline{W_n}$ can be the same.

- (b) The proof is similar to the previous one, only the beginning and ending four terms of W_n and $\overline{W_n}$ must be considered. By a similar argument to the previous part, the beginning and ending four terms of any W_n or $\overline{W_n}$ are either 0110 or 1001. Starting with $W_3 = 01101001$ and $\overline{W_3} = 10010110$, neither of which contain 01010 or 10101, the induction now proceeds as above. The desired subwords cannot occur completely with the W_n or $\overline{W_n}$ part of W_{n+1} by the inductive assumption, and they cannot be “in the middle” as the eight terms in the middle must be 01100110, 01101001, 10010110, or 10011001, none of which contain the desired subwords.

The graders saw several much nicer proofs of these last two statements. Here are a few:

- (a)
- From problem (9), the W_n and $\overline{W_n}$ are constructed from the μ operation, which always produces different entries in two consecutive terms. Thus, since any three consecutive terms must contain two that come from the same μ operation, there can be no three consecutive equal terms.
 - Similarly, for any three consecutive nonnegative integers, either the first or second must be even, and then the second or third, respectively, will be the following odd number, and these have different parities of numbers of 1's in their binary expansions.
 - Any W_k or $\overline{W_k}$ is composed entirely of $W_1 = 01$ and $\overline{W_1} = 10$. But the four possible combinations of these are 0101, 0110, 1001, and 1010, none of which contain three consecutive equal entries.
- (b)
- Using the μ operation, the string 01010 starting in position $2k$ had to come from 000 in position k , while 01010 in position $2k + 1$ came from 111 in position k . Similarly, 10101 had to come from either 000 or 111. Since, by part (a) no triple repetition ever occurs, neither can 10101 or 01010.
 - Any W_k or $\overline{W_k}$ is made up of 1001 or 0110 (or is too short to contain the given five-digit sequences). But the combinations of these are 10011001, 10010110, 01101001, and 01100110, none of which contain the desired subwords.